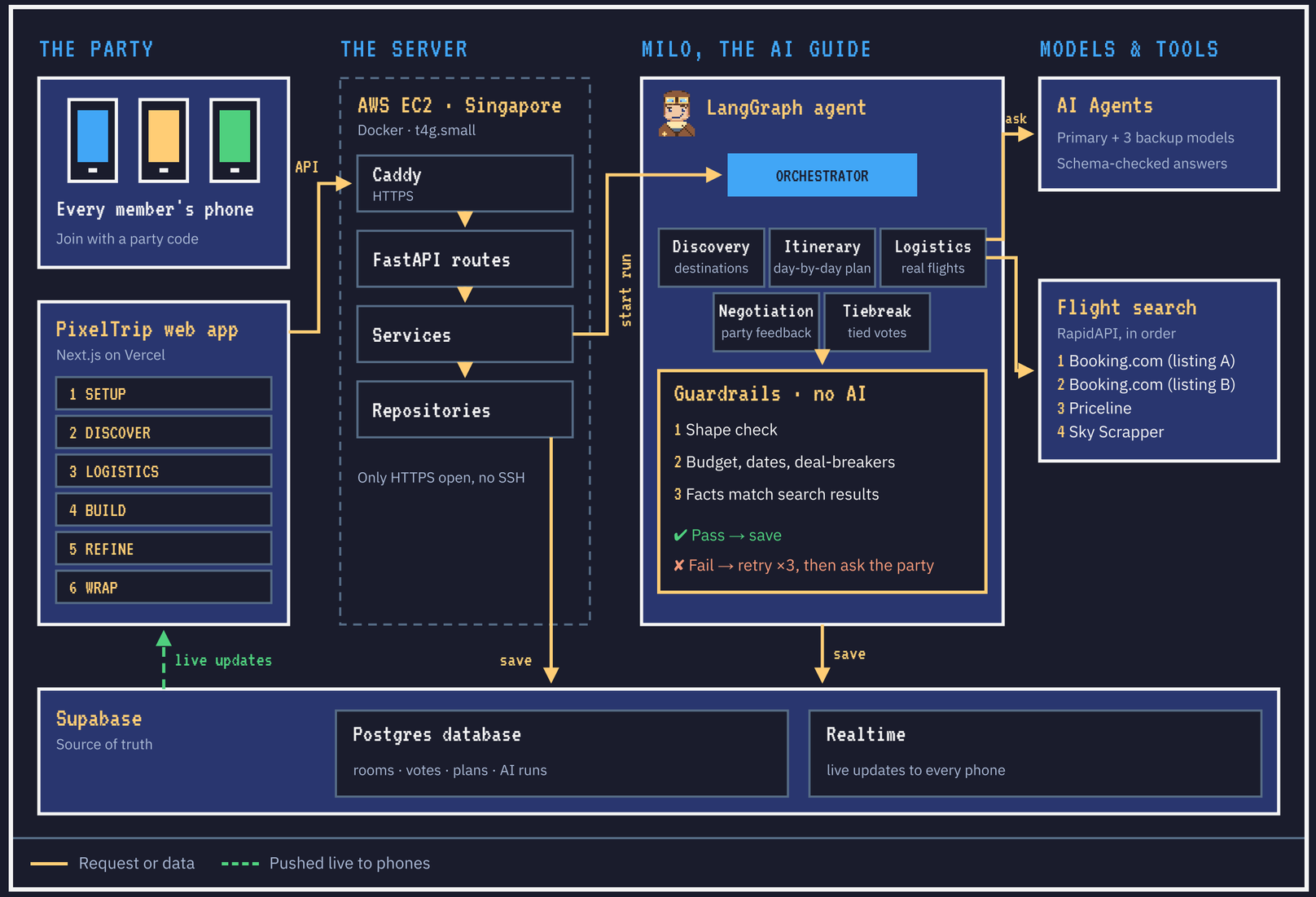
PixelTrip Architecture

MILO suggests. Code checks. The party decides.



The trip pipeline

1 Setup — Group profile
Party makes pixel characters

2 Discover — Discovery · Tiebreak
MILO suggests places → party votes

3 Logistics — Logistics
MILO finds real flights → party votes

4 Build — Itinerary
MILO drafts the day-by-day plan

5 Refine — Negotiation
Party flags changes → MILO redrafts → host approves

6 Wrap — Export
Party saves the plan as Markdown, text or calendar

Deployment

FRONTEND: git push → GitHub → Vercel build → Live site

BACKEND: deploy.sh → Docker ARM image → Amazon ECR → SSM Run Command → EC2 pulls & restarts → /health check

Parameter Store holds the keys, never the repo · Caddy gets the HTTPS certificate

More than an LLM wrapper

960+ automated tests
812 backend, 155 frontend, incl. property-based

3 guardrail layers
Shape, rules and facts, checked in plain Python

0 invented flights
Flights and prices must match search results exactly

Only what changed
Refine redrafts flagged days, the rest stays as is

8 fallbacks
4 AI models and 4 flight providers, tried in order

Guarded party input
Members' text can't give MILO instructions